

SecurePouch PCB Electrical Review Specification

nRF52840 + nRF9151 + nPM1300 smart pouch PCB

Status	Electrical review draft
Last updated	June 2026
Primary schematic	smart_pouch.kicad_sch
Purpose	Reviewer checklist for power, USB-C, battery, PMIC, and high-current loads

1 Review Scope

This document is intentionally limited to electrical design information needed for schematic review. Mechanical, backend, app, and product-roadmap details have been removed from this review copy.

2 System Summary

Block	Schematic ref.	Notes
BLE/application MCU	U1, nRF52840	Local peripherals, BLE, USB data, PMIC control over I2C/TWI
Cellular/GNSS SiP	U2, nRF9151-SICA	LTE-M/NB-IoT/GNSS; UART link to nRF52840
PMIC/charger	U?, nPM1300-CAAA-R7	USB-C input, battery charging, BUCK1/BUCK2, fuel/ADC monitoring
Battery connector	CN1	Intended 3-pin, 2.54 mm battery connector; pinout: VBAT+, NTC, GND
USB-C receptacle	USB1, C2765186	16-pin USB2.0 Type-C receptacle
Accelerometer	BMA400	I2C, interrupt lines to nRF52840
Siren boost	MT3608	15 V rail for piezo H-bridge
Solenoid boost	MT3608	12 V rail for lock solenoid pulse

3 Power Architecture

Rail/net	Source	Nominal	Consumers / review notes
VBAT	1S LiPo via CN1	3.0–4.2 V typical	nPM1300 VBAT. Battery pack must include PCM/protection.
VSYS	nPM1300 power path	USB or battery dependent	System input rail. Do not drive VSYS from an external source.
+1V8	nPM1300 BUCK1	1.8 V	nRF52840 core/I/O domain and low-voltage logic as drawn.
+3.3V	nPM1300 BUCK2	3.3 V target	Application 3.3 V loads. VSET2 resistor must select 3.3 V at power-up.
+12V	MT3608 boost	12 V	Solenoid only. Pulse load; verify diode/MOSFET stress and peak current.
+15V	MT3608 boost	15 V	Piezo H-bridge only. Produces 30 Vpp differential drive.
VBUS	USB-C	5 V nominal	nPM1300 VBUS input. Add local VBUS decoupling and USB ESD protection.

3.1 nPM1300 Notes

- nPM1300 VBUS is the charger/system input. VBUSOUT is not required for charging.

- VBUSOUT is only a protected VBUS sense output for a host USB VBUS detect input. Leave NC if unused.
- Battery charging is firmware-enabled over TWI/I2C; configure charge current, termination voltage, NTC type, and VBUS current limit before production use.
- BUCK outputs must not be tied together.
- nPM1300 recommended BUCK external components from datasheet: 2.2 uH inductor, effective output capacitance at least 4 uF, low ESR.
- VSET pins are sampled at power-on only. Do not leave VSET1/VSET2 floating.

3.2 BUCK Startup Configuration

Pin	Target	Required resistor to GND	Review note
VSET1	1.8 V	47 k Ω	Verify against schematic/BOM.
VSET2	3.3 V	250–500 k Ω	Use 330 k Ω nominal unless BOM already has another value in range.

4 Battery and NTC

CN1 pin	Net	Function
1	VBAT	Battery positive
2	NTC	10 k Ω NTC to battery negative
3	GND	Battery negative / system ground

CN1 is intended to be a 3-pin, 2.54 mm battery connector. Do not label this as JST-PH unless the selected part is actually PH-series; JST-PH is 2.0 mm pitch. If a 2.54 mm connector is used, identify the exact family/part number in the schematic and BOM. Silkscreen should identify the battery pinout as + NTC -. Confirm footprint pin order against the mating cable before ordering.

NTC requirements:

- Preferred pack: 1S LiPo, approximately 1000 mAh, protected, with 10 k Ω NTC.
- nPM1300 default temperature thresholds are intended for a 10 k Ω NTC profile.
- If no NTC is used, the nPM1300 NTC function must be disabled in firmware and the NTC pin tied as specified by the datasheet. For this product, using a real battery NTC is preferred.

5 USB-C Interface

USB-C pins	Net	Destination	Notes
A4/A9/B4/B9	VBUS	nPM1300 VBUS	Tie all VBUS contacts together. Local VBUS decoupling required.
A1/B12/B1/A12	GND	Board GND	Tie all ground contacts together.
A5	CC1	nPM1300 CC1	Connect directly; nPM1300 provides Rd/current detection.
B5	CC2	nPM1300 CC2	Connect directly; nPM1300 provides Rd/current detection.
A6/B6	USB_D+	nRF52840 USB D+	Tie connector D+ contacts together. Use ESD protection.
A7/B7	USB_D-	nRF52840 USB D-	Tie connector D- contacts together. Use ESD protection.
A8/B8	SBU1/SBU2	NC	Unused for USB2 charging/data.
13/14	Shield/EH	Chassis/GND strategy	Connector shell pads. Prefer direct GND or defined shield network near connector.

Important: Do not add external 5.1 k Ω CC pulldowns when CC1/CC2 are connected to the nPM1300; the nPM1300 implements the Type-C sink detection path internally.

6 User Button, Ship Mode, and Reset

Signal	Connection	Notes
SHPHLD	User button to GND	Wakes nPM1300 from Ship/Hibernate. Internal pull-up to VBAT/VBUS.
NPM_INT	nPM1300 GPIO1 to nRF52840	Used for PMIC events, including SHPHLD events if configured.
NPM_RESET	nRF52840 to nPM1300 GPIO0	MCU-controlled reset/control path. Keep separate from the user button.

Recommended user-button behavior:

- Off/Ship: button pulls SHPHLD low and wakes the nPM1300.
- On: nPM1300 reports button event to nRF52840 through PMIC interrupt and TWI status registers.
- Firmware long-press action: nRF52840 commands nPM1300 to enter Ship mode.
- nPM1300 long SHPHLD press reset is enabled by default; decide whether firmware should disable or keep it.

7 High-Current / High-Voltage Loads

Load	Rail	Driver	Control	Review notes
Solenoid lock	+12V	2N7002 low-side	nRF52840 GPIO	Verify solenoid current, MOSFET pulsed SOA, flyback path, and pulse duration.
Piezo siren	+15V	MOSFET H-bridge	2.9 kHz complementary PWM	Verify no shoot-through, gate resistor values, and piezo voltage limited to 30 V _{pp} .
Strobe LED	+3.3V	MOSFET low-side	PWM/strobe GPIO	Verify current limit topology, peak current, thermal rise, and BUCK2 current budget.

8 Schematic Cross-Reference

9 Open Review Items

1. Confirm nPM1300 package choice and assembly capability. QFN is preferred for prototype inspectability; WLCSP is acceptable only with proven assembler capability.
2. Confirm all nPM1300 required decoupling capacitors and BUCK passives match the datasheet/reference design.
3. Resolve battery connector naming: 2.54 mm is acceptable if it matches the cable, but it should not be called JST-PH unless it is a real 2.0 mm PH-series part.
4. Confirm USB-C shield grounding strategy and ESD placement before layout.
5. Confirm charge profile for the selected cell: chemistry, VTERM, ICHG, JEITA thresholds, and NTC beta selection.

6. Confirm +3.3V BUCK2 load budget. nPM1300 BUCK outputs are 200 mA class rails; strobe LEDs may need a separate driver/rail if peak current exceeds budget.
7. Confirm +12V solenoid and +15V siren boost converters are not enabled during low battery, brownout, or USB current-limit states without firmware gating.
8. Confirm all nets driven from VBAT/VBUS into nRF52840 pins are level-safe. SHPHLD is not safe to tie directly to a 3.3 V-only GPIO.
9. Run KiCad ERC after connector pin types are cleaned: connector pins passive, USB shell passive, explicit PWR_FLAGS only where needed.

Item	Current schematic evidence	Review action
USB-C connector	USB1, C2765186, 16-pin USB2 receptacle	Confirm shield pins 13/14 are passive and connected intentionally.
USB CC routing	CC1/CC2 labels present at USB1 and nPM1300	Confirm no external CC Rd resistors are also fitted.
USB data	USB_D+ / USB_D- labels present at USB1 and nRF52840	Confirm D+/D-polarity and ESD part orientation.
Battery connector	CN1 intended as 2.54 mm 3-pin	Confirm exact connector family/part number and pin order 1=VBAT, 2=NTC, 3=GND against footprint/cable.
Battery section text	Schematic text says “Battery JST 2.54mm 1x3P”	Keep 2.54 mm if that is the selected pitch, but remove “JST-PH” wording unless using a real PH-series 2.0 mm connector.
VSET2	VSET2 net present on nPM1300	Confirm resistor value is 250–500 k Ω for 3.3 V startup.
VBUSOUT	nPM1300 VBUSOUT pin present	If NC, no capacitor required; if used for host VBUS sense, add required decoupling.
SHPHLD	SW1 near nPM1300; SHPHLD pin present	Confirm SW1 pulls SHPHLD to GND only, not directly to nRF GPIO at VBAT voltage.
PMIC interrupt	NPM_INT label between nPM1300 and nRF52840	Confirm firmware will enable SHPHLD/button events if button has no direct MCU GPIO.